PERIODIC REPORT

HIGH PERFORMANCE COMPUTING

#XX

Reporting Period: 25_MMDD - 25_MMDD

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Meta Information

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Main Sheet:	**Google Sheet**
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References Sheet:	**Google Sheet**

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Project Overview

- **LoI:** lmao ngu
- **Status:** [DONE] C++/Python templates done, [WIP] LaTeX in progress.
- **Tools:** LuaLaTeX, latexmk, Beamer, TiKZ.

Section Contents

Mathematical Foundations

Euler's Identity

$$e^{i\pi} + 1 = 0$$

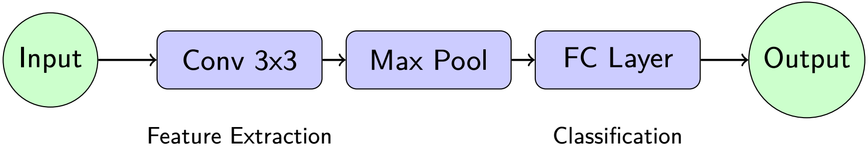
Schrödinger Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[\frac{-\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t)$$

Matrix operations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \times \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

Deep Learning Architecture (TiKZ)



Algorithm Implementation

Algorithm 1: Min-Max Normalization

Data: Input image I

Result: Normalized image I'

$min, max \leftarrow$ compute_range(I);

for *each pixel* $p \in I$ **do**

  $I'(p) \leftarrow \frac{I(p) - min}{max - min} \times 255;$

end

Split Layout Examples

Left Column (Blocks)

Standard 50/50 split using:

- `columns` environment
- `column` environment

This is great for comparing text with images or code.

Right Column (Info)

You can place anything here:

- Tables
- Graphics
- Equations

Asymmetric Split (60/40)

```
def analyze_data(data):
    # Complex math here
    result = [x**2 for x in data]
    return sum(result)

```

Processing Logic

Observation

The left column takes 60% width for code, while this alert takes 35% for explanation.

Source Code (C++ & Python)

main.cpp

```cpp
#include <iostream>

int main() {
    // Hello Panda!
    std::cout << "Hi!" << std::endl;
    return 0;
}
```

```python
import numpy as np

def hello():
    print("Hello from Python!")
    return np.zeros(5)

```

Code: Python Snippet

Section Contents

Citation Gallery

- Foundational TeX: [1]
- LaTeX Guide: [2], [3]
- Literate Programming: [4]
- Research Examples: [5], [6]

Full Citation Example

Featured Reference

D. E. Knuth, *The TEX Book*. Addison-Wesley Professional, 1986

Another Full Citation

F. Mittelbach et al., *The LATEX Companion*, 2nd ed. Addison-Wesley Professional, 2004

THANK YOU

for your attention

References I

[1] D. E. Knuth, *The TEX Book*. Addison-Wesley Professional, 1986.

[2] L. Lamport, *LATEX: a Document Preparation System*, 2nd ed. Massachusetts: Addison Wesley, 1994.

[3] F. Mittelbach, M. Gossens, J. Braams, D. Carlisle, and C. Rowley, *The LATEX Companion*, 2nd ed. Addison-Wesley Professional, 2004.

[4] D. E. Knuth, "Literate programming," *The Computer Journal*, vol. 27, no. 2, pp. 97–111, 1984.

[5] G. Foster, S. Gandrabur, P. Langlais, P. Plamondon, G. Russell, and M. Simard, "Statistical machine translation: Rapid development with limited resources," in *Proceedings of MT Summit IX*, New Orleans, USA, 2003, pp. 110–119.

[6] E. Alsolami, "An examination of keystroke dynamics for continuous user authentication," Ph.D. dissertation, Queensland University of Technology, 2012.